

CAREER141

CV Processing & Recruitment Pipeline

Project Proposal & System Documentation

Executive Summary

Career141 proposes a fully automated, AI-powered recruitment pipeline designed to eliminate manual effort, reduce time-to-hire, and surface the best candidates faster. This system handles the complete lifecycle of a candidate — from initial CV intake through 7 ingestion channels, AI-powered parsing, semantic matching across 115,000+ profiles, multi-channel proactive outreach, multi-stakeholder review, and final placement.

8 specialised AI agents operate autonomously across the pipeline, handling tasks that would otherwise require dozens of manual hours per week — from monitoring WhatsApp and email inboxes 24/7, to deduplicating candidate profiles, to running proactive outreach campaigns and daily pipeline health checks.

115,000+

Candidate Profiles

8

AI Agents

7

Ingestion Channels

13

Pipeline Stages

Prepared: June 2026

Version: 1.0 | Career141 Internal

1. System Overview

Career141 operates a fully automated, AI-powered recruitment pipeline that handles the complete lifecycle of a candidate — from the moment their CV arrives through any one of 7 ingestion channels, all the way through parsing, matching, outreach, multi-stakeholder review, interview, and final placement.

Platform Highlights
• Processes 50–600+ CVs per bulk upload with continuous multi-channel intake
• Manages a searchable database of 115,000+ candidate profiles
• 8 specialised AI agents handle pipeline tasks autonomously, 24/7
• Multi-stakeholder review workflow: Talent Acquisition → Director → Client
• Full audit trail — every candidate action, communication, and stage change is logged
• Hot/cold two-tier storage keeps operational costs low while keeping all records accessible

Core Technology Stack

Component	Technology	Purpose
Ingestion Gateway	AWS API Gateway	Central entry point for all CV submissions
Application Routing	Node.js Express	Routes CVs to correct processing pipelines
Hot Storage	Convex DB	30-day buffer for active/recent candidates
Cold Archive	MinIO	Long-term self-hosted S3-compatible archive
AI Parsing Model	OpenRouter Nemotron 30B	Structured field extraction from CVs (current)
Deduplication	SHA-256 Hashing	Exact-match duplicate file detection
Semantic Search	Vector Embeddings	Similarity-based candidate discovery
Text Extraction	pdfjs-dist / mammoth	Extract raw text from PDF and Word files

2. System Flow — 13 Stages

Every CV that enters the Career141 system passes through the same 13-stage pipeline, regardless of its source channel. The pipeline is fully automated from Stage 1 through Stage 7, with human decision-making introduced at Stage 11 (recruiter shortlisting) and maintained through placement.

Stage 1	CV Sources — 7 Ingestion Channels
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What Happens: CVs enter the system simultaneously from 7 distinct channels. Every channel feeds the same unified pipeline, but each source is tagged independently for attribution tracking. Candidates are only routed into a job pipeline if that job is currently in **Active** status — if a job is On Hold, Filled, or Cancelled, incoming CVs are still captured and stored in the candidate database but are not added to that job's pipeline. This prevents candidates from being accidentally assigned to roles that are no longer open.

Why It Matters: Candidates apply through different means — active seekers use LinkedIn, while passive candidates are reached via headhunting or Meta Campaigns. Capturing all channels ensures no candidate is missed, while Active-only routing ensures every candidate lands in a live, working pipeline rather than a completed or paused one.

- **LinkedIn (Unified Email Intake)** — Easy Apply, External, Lead Gen Forms, and Headhunting all route CVs to a single shared inbox (e.g. linkedin@career141.com) with the job keyword in the subject line
- **WhatsApp** — job-specific phone numbers; candidates send CV as file attachment
- **Meta Campaign** — shared WhatsApp line driven by paid Facebook/Instagram ads
- **Email Campaign** — dedicated inbox per job (e.g. fin2024@career141.com)
- **Workable ATS** — webhook sync when a new application is received in Workable
- **Manual / Bulk Upload** — TA uploads 1 to 600+ CVs directly, selecting the target job
- **Headhunting** — TA-sourced passive candidates added manually and tagged
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Stage 2

Keyword / Job ID Routing Setup

What Happens: Before CVs can be routed, each job opening is configured with a unique keyword, a shared LinkedIn intake address, and dedicated channel-specific identifiers.

Why It Matters: Without routing setup, the system cannot know whether a WhatsApp CV belongs to a Finance Manager or Software Engineer role. The keyword is the address label on every incoming CV.

- A unique job keyword (e.g. FIN2024, ENGSR) is assigned at job creation
- A single shared LinkedIn intake email receives all LinkedIn sub-channel CVs — job keyword in the subject line routes them correctly
- Dedicated email inbox per Email Campaign job
- Dedicated WhatsApp number per job (where applicable)
- Routing rules map all identifiers to the correct job record in the database

Stage 3

Channel Routing Agents

What Happens: Automated bots and monitors watch each channel continuously. When a CV arrives, the routing agent extracts it, identifies the job keyword or inbox address, and pushes the CV — fully tagged — to the API Gateway.

Why It Matters: Each channel speaks a different language. These channel-specific agents act as translators and normalise all inputs before the central pipeline processes them.

- LinkedIn Email Monitor: reads email subject line for job keyword, tags LinkedIn sub-channel, forwards to API Gateway
- WhatsApp Bot (Agent 4): 24/7 monitor; identifies job from WhatsApp number; extracts CV attachment

- Email Monitor (Agent 7): watches all job-specific campaign inboxes live; routes by inbox address
- Meta/WhatsApp Campaign Router: keyword detection in conversation to identify job
- Workable Webhook Listener: maps Workable job ID to C141 job ID
- Manual Upload Processor: applies job ID selected by TA at upload time

Stage 4**Central Ingestion via AWS API Gateway**

What Happens: All CVs from all channels converge at the AWS API Gateway — the single front door of the system. Every CV is authenticated, validated, tagged, and passed to the Node.js router.

Why It Matters: Centralising all ingestion at one point means one place to monitor, log, and manage all incoming CVs, with consistent handling regardless of source.

- AWS API Gateway receives the POST request with CV file and metadata
- Request is authenticated (API key / token check)
- Payload validated: CV file present, job ID attached, source tagged
- Node.js Express receives validated request and applies routing logic
- A unique record ID is generated for the submission

Stage 5**Storage Architecture — Two-Tier System**

What Happens: On arrival, every CV is simultaneously stored and checked for duplicates using SHA-256 hashing. New CVs go into Convex DB (hot storage); older records are archived to MinIO (cold storage).

Why It Matters: Prevents duplicate processing and provides a cost-efficient storage strategy — fast access for recent CVs, cheap archival for historical records.

- SHA-256 hash fingerprints every CV file — exact duplicates are caught immediately without AI processing
- Duplicate found: existing record updated with new source tag; no new record created
- New CV: written to Convex DB (30-day hot storage, real-time access)
- After 30 days: migrated to MinIO (self-hosted S3-compatible cold archive)
- `isArchivedLocally` flag tracks archival status; archived records fetched from MinIO transparently when accessed

Stage 6**Lazy CV Parsing & AI Extraction — Agent 1**

What Happens: Parsing is triggered on first recruiter access, not at ingestion. When opened, Agent 1 extracts 13+ structured fields using AI and assigns confidence scores to each.

Why It Matters: Parsing every CV at ingestion is expensive, especially for bulk uploads of 600+ CVs. Lazy parsing means AI compute is only spent on CVs that are actually viewed — significantly reducing costs.

- Triggered on first TA access to a candidate profile
- Text extracted from PDF (pdfjs-dist) or Word doc (mammoth)
- AI model extracts: Name, Email, Phone, Location, Job Title, Employer, Experience, Skills, Education (Degree / Institution / Year), Languages, LinkedIn URL

- Each field receives a confidence score (0–1.0); low-confidence fields flagged for human review
- Skill normalisation applied: 'JS' → 'JavaScript', 'ML' → 'Machine Learning'
- Current model: OpenRouter Nemotron 30B at temperature=0 for deterministic outputs

Stage 7**Deduplication & Profile Creation — Agent 6**

What Happens: After AI extraction, Agent 6 performs deep multi-factor deduplication, merges duplicate profiles, creates the final candidate record, and generates vector embeddings for semantic search.

Why It Matters: The SHA-256 check catches identical files. Agent 6 catches near-duplicates — the same person applying from two channels or submitting a slightly updated CV.

- 4-factor duplicate detection: exact email match, exact phone match, LinkedIn URL match, fuzzy name match
- Duplicate confirmed: profiles merged; new application added as event on existing profile; both CVs retained in history
- New candidate: complete profile created with all extracted and validated fields
- Vector embeddings generated from full profile text for semantic search capability
- Candidate goes LIVE and becomes searchable across the full 115,000+ database

Stage 8**Job Dashboard — Source-Categorised View**

What Happens: Candidates appear in recruiter dashboards organised by job opening and source channel. Every candidate has an AI match score (0–100) showing fit against the role requirements.

Why It Matters: Recruiters handle multiple jobs across many channels. The dashboard gives a clean, prioritised view — highest-match shortlisted candidates first, with source visibility for attribution tracking.

- All active jobs listed; each job shows Short listed candidates grouped into tabs by source channel
- AI match score (0–100) calculated by Agent 2 for every candidate against the job
- Candidates sorted by match score by default (highest first)
- Score factors: skills overlap, years of experience, job title relevance, industry background, location
- Visual indicators for candidates with low-confidence parsed fields requiring review

Stage 9**CV Search — Semantic Search Across Full Database — Agent 2**

What Happens: Agent 2 enables freeform semantic search across all 115,000+ candidate profiles. Recruiters enter a natural language query or paste a job description and get ranked results.

Why It Matters: The best candidate for a new role may have applied 18 months ago. CV Search turns the entire historical database into an active talent pool for every new job opening.

- Recruiter enters natural language query or job description
- Query converted to vector embedding and compared against all 115k+ candidate vectors
- Top matches retrieved and re-ranked; each scored 0–100 against job requirements

- Supports semantic search (concept-based), keyword/filter search (hard criteria), and hybrid mode
- Filters: location, experience, skills, languages, availability

Stage 10**AI Proactive Outreach — Agent 3 & Agent 5**

What Happens: The system proactively reaches out to candidates via email follow-up sequences and AI phone calls — covering both new applicants and candidates identified from the 115,000+ database.

Why It Matters: Many of the best candidates are passive — not actively applying but open to the right opportunity. Proactive outreach surfaces these candidates before they appear on any competitor's radar.

- Agent 3 (Follow-Up): Day 2 initial email, Day 4 follow-up email, Day 7 final email — stops immediately on reply or opt-out
- Agent 5 (AI Phone Call): calls both new applicants and existing database candidates matched to the role
- IVR — Press 1 to express interest, Press 2 to decline
- Press 1: candidate flagged as Interested and added to headhunting pipeline for TA follow-up
- Press 2: marked Not Interested for this role; remains in DB for future roles
- All outreach events, responses, and outcomes logged to candidate profile timeline

Stage 11**Recruiter Shortlist & Multi-Level Review**

What Happens: After AI scoring and proactive outreach, human decision-making begins across three sequential review levels. Each level has its own dedicated tab within the job view, and within each tab, shortlisted candidates are organised into sub-tabs by the source channel they came from — making it easy to evaluate candidate quality per channel at every stage of review.

Why It Matters: A three-level review process prevents unqualified CVs from reaching clients, protects Career141's reputation, and gives clients a pre-vetted shortlist they can trust. Organising candidates by source within each review tab gives recruiters and directors instant visibility into which channels are producing the strongest candidates for each specific role.

- **Level 1 — TA Recruiter Tab:** The recruiter reviews all AI-ranked candidates and selects their shortlist. Shortlisted candidates are displayed within sub-tabs by source channel — LinkedIn, WhatsApp, Email Marketing, and Others. The recruiter adds personal notes and records rejection reasons for any candidates not carried forward.
- **Level 2 — Director Tab:** The director reviews the TA's approved shortlist, again organised by source sub-tabs — LinkedIn, WhatsApp, Email Marketing, and Others. The director can approve candidates to pass to the client, request changes, or reject candidates with logged feedback.
- **Level 3 — Client Tab:** The client reviews the director-approved shortlist, presented in the same source sub-tab structure — LinkedIn, WhatsApp, Email Marketing, and Others. The client marks each candidate as Selected, Hold, or Rejected.
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- Only 'Selected for Interview' candidates proceed to Stage 12
- All decisions and timestamps recorded in full audit trail

Stage 12 Interview → Offer → Placement

What Happens: Client-selected candidates enter interview coordination. Interviews are scheduled, feedback captured, offers made, and outcomes recorded — placement or return to searchable database.

Why It Matters: Closing the loop on every candidate ensures the database reflects current availability and that no placement or rejection is ever lost from the audit trail.

- Client schedules interview; system coordinates and logs date/time
- Post-interview: client submits structured feedback (rating + comments) against candidate profile
- Offer extended: offer details (salary, start date, terms) recorded; status changed to Offer Extended
- Accept: status set to Placed; pipeline notification sent; placement recorded for billing/reporting
- Reject / Withdraw: status updated to Available; candidate remains searchable for future roles

Stage 13 Candidate Profile & Complete History

What Happens: Every candidate has a permanent, comprehensive profile that accumulates their full journey — regardless of outcome. Profiles remain active and searchable forever.

Why It Matters: A candidate rejected today may be the perfect fit 12 months later. Permanent searchable profiles mean the 115,000+ database grows in value with every passing month.

- All CV versions ever submitted (with date and source channel)
- Every stage reached, with timestamps and decision-maker identity
- All communications: emails, WhatsApp messages, phone call outcomes, automated follow-ups
- Match scores for every job ever considered
- Interview feedback and placement history (where applicable)
- Low-confidence AI extraction flags for human-verified fields

3. The 8 AI Agents

The C141 Platform coordinates 8 specialised AI agents that automate distinct parts of the recruitment pipeline. Each agent has a specific trigger, a defined scope of action, and a clear output. Together they eliminate the need for manual monitoring, manual outreach, and manual deduplication.

Agent	Name	Trigger	Key Function
Agent 1	CV Parsing Agent	On first profile view	Extracts 13+ structured fields from raw CVs with confidence scoring
Agent 2	Matching Agent	On job open / search	Scans 115k+ candidates, ranks and scores 0–100 against job requirements
Agent 3	Follow-Up Agent	On outreach initiation	Day 2/4/7 email sequence to engage unresponsive candidates

Agent	Name	Trigger	Key Function
Agent 4	WhatsApp Monitor	Continuous (24/7)	Watches job WhatsApp numbers, detects CV attachments, routes to pipeline
Agent 5	AI Phone Call Agent	On proactive outreach	Calls new applicants and database candidates; captures interest via IVR (Press 1/2), routes to TA
Agent 6	Deduplication Agent	On new CV submission	4-factor dedup (email/phone/LinkedIn/fuzzy name), profile merge/creation
Agent 7	Email Monitor Agent	Continuous (live)	Watches LinkedIn common inbox + job-specific campaign inboxes; routes CVs
Agent 8	Pipeline Health Agent	Daily scheduled scan	Flags stalled jobs, nudges responsible staff, generates health report

Agent 1**CV Parsing Agent***Trigger: First recruiter access to a candidate profile (lazy trigger)*

- Receives raw CV file (PDF or Word) and extracts plain text
- Sends extracted text to AI language model with structured extraction prompt
- Extracts 13+ fields: Name, Email, Phone, Location, Job Title, Employer, Experience, Skills, Education, Languages, LinkedIn URL
- Assigns confidence score (0–1.0) to every extracted field
- Fields below confidence threshold flagged for human review
- Applies skill normalisation (e.g. JS → JavaScript, ML → Machine Learning)
- Model temperature = 0 for deterministic, consistent extractions
- Current model: OpenRouter Nemotron 30B | Recommended upgrade: Claude Sonnet 4.6 or Claude Opus 4.7

Agent 2**Matching Agent***Trigger: New job opened or recruiter runs a search*

- Takes job description or search query as input
- Converts query to vector embedding using the configured embedding model
- Performs semantic similarity search across all 115,000+ candidate embeddings
- Retrieves top N candidates (e.g. top 200) and calculates detailed match score (0–100) for each
- Score factors: skills overlap, experience relevance, job title similarity, industry background, location fit
- Supports semantic (natural language), keyword/filter, and hybrid search modes
- Results delivered to Job Dashboard (auto-scored) and CV Search (on-demand)

Agent 3**Follow-Up Agent***Trigger: Candidate identified for outreach or manual trigger by TA*

- Sends structured multi-day follow-up email sequence to unresponsive matched candidates
- Day 2: initial outreach email about the opportunity
- Day 4: follow-up email if no response

- Day 7: final follow-up email
- Sequence stops immediately on candidate reply, opt-out, or progression in pipeline
- All messages personalised with candidate name and role details
- All send, delivery, and read events logged to candidate communication timeline

Agent 4**WhatsApp Monitor Agent***Trigger: Continuous — runs 24/7 in real time*

- Monitors all active job-specific WhatsApp numbers simultaneously
- Detects incoming messages and scans for CV file attachments (PDF, Word)
- Identifies the job from the WhatsApp number the message was sent to
- Extracts CV file, attaches job ID and source tag 'WhatsApp', pushes to API Gateway
- If job cannot be determined: sends automated reply asking candidate to confirm the role
- Captures candidate phone number and name from WhatsApp profile metadata

Agent 5**AI Phone Call Agent***Trigger: New applicants entering the pipeline and database candidates matched to a role*

- Calls two groups: new applicants who have entered the pipeline via any ingestion channel, and existing database candidates identified as a strong match for a new role
- Initiates outbound call to candidate's phone number
- Plays automated message introducing the opportunity (role name, company type, key details)
- Interactive voice response: Press 1 = Interested, Press 2 = Not Interested
- Press 1: candidate flagged as 'Interested — Awaiting TA Follow-Up'; added to headhunting pipeline; TA notified
- Press 2: marked 'Not Interested — This Role'; sequence stops; candidate remains in DB for future roles
- No answer / voicemail: logged as 'No Answer'; Agent 3 follow-up email sequence may activate
- All call outcomes, timestamps, and recordings stored in candidate profile

Agent 6**Deduplication Agent***Trigger: Every new CV submission after initial SHA-256 check*

- 4-factor duplicate detection: exact email, exact phone, LinkedIn URL, fuzzy name match
- Duplicate confirmed: profiles merged; new application added as event on existing profile; both CVs retained
- New candidate: complete profile assembled from all extracted and validated fields
- Generates vector embeddings for the profile (enables semantic search)
- Writes live, searchable profile to database

Agent 7**Email Monitor Agent***Trigger: Continuous — runs live in real time*

- Monitors two inbox types simultaneously

- LinkedIn Shared Inbox (linkedin@career141.com): reads email subject line for job keyword as routing signal; tags LinkedIn sub-channel from metadata
- Job-Specific Campaign Inboxes (e.g. fin2024@career141.com): routes by inbox address; no keyword reading needed
- Extracts CV attachment, attaches job ID and source tag, pushes to API Gateway
- Captures candidate email address from sender field
- Handles edge cases: multiple attachments, forwarded emails, unusual formats
- Sends automated confirmation reply to candidate acknowledging receipt

Agent 8**Pipeline Health Agent***Trigger: Daily scheduled scan (e.g. 8:00 AM every morning)*

- Scans all active jobs and current pipeline status each morning
- Flags 'On Hold' jobs open longer than expected SLA without progression
- Flags candidates stuck at a stage (e.g. Director Review) for too long without decision
- Detects jobs with no new CV activity in X days (possible sourcing issue)
- Generates daily health report across all active roles
- Sends nudge notifications to responsible recruiter or director
- Tracks pipeline metrics over time: time-to-shortlist, time-to-client, time-to-placement
- Alerts on anomalies: sudden CV volume drops, parsing failure rates

4. Key System Features

SHA-256 SHA-256 File-Level Deduplication

Every CV file is fingerprinted with a SHA-256 hash at ingestion. Identical files are caught immediately — before any AI processing occurs — saving compute costs and preventing duplicate records.

Lazy Lazy CV Parsing

AI extraction fires only when a recruiter opens a candidate profile. For bulk uploads of 600 CVs, the system ingests and stores all 600 instantly — but only parses the ones actually viewed. This significantly reduces AI costs on high-volume batches.

0–1.0 Confidence-Scored AI Extraction

Every field extracted by Agent 1 carries a confidence score (0–1.0). The system knows when it is uncertain, and flags those fields for human verification — preventing silent errors in the candidate database.

Vector Vector Semantic Search

The entire 115,000+ candidate database is indexed with vector embeddings. This enables search that understands meaning — 'Senior finance professional with banking experience' finds candidates whose CVs say 'VP Treasury & Capital Markets'.

Source Multi-Channel Source Attribution

Every candidate record permanently tracks their source channel, submission date, and the job they applied for. This powers channel effectiveness reporting — e.g. 'LinkedIn generates 40% of placements but only 20% of applications'.

Hot/Cold Two-Tier Hot/Cold Storage

Convex DB provides 30-day fast, real-time access for active candidates. MinIO provides cheap long-term archival. The `isArchivedLocally` flag ensures transparent access — recruiters never notice the storage tier.

Audit Multi-Stakeholder Approval Workflow

TA Recruiter → Director → Client review chain ensures quality at every gate. Every approval, rejection, and piece of feedback is timestamped and attributed — full audit trail from CV submission to placement.

AI Proactive AI Outreach

The system reaches out to both new applicants and historical database candidates matched to open roles. Agent 3 sends structured email follow-up sequences; Agent 5 makes AI phone calls to both groups — turning a static archive into an active talent network.

Trail Full Audit Trail

Every action on every candidate is timestamped and logged — who viewed them, what stage they moved to, what communications were sent, what feedback was given. Complete traceability from first submission to final placement.

5. AI Model Recommendations

Each agent in the pipeline requires different AI capabilities. The tables below show recommended models for each role — from the current lightweight implementation to enterprise-grade options.

5.1 — Agent 1: CV Parsing — Vision Language Models

Model	Provider	Strength
Google Document AI	Google Cloud	Enterprise-ready; high-volume document processing with Gemini-powered multimodal reasoning
LlamaIndex LlamaParse	LlamaIndex	Handles tables, charts, images, handwriting, and multi-column layouts; agentic self-correction
Doxis AI.dp	Doxis	>99% accuracy with real-time results; GDPR compliant; strong system integration support
ABBYY FlexiCapture	ABBYY	Strong enterprise-grade capture for large-scale document processing projects
Amazon Textract	AWS	AWS-native; multi-language support; integrates directly with the existing API Gateway infrastructure

5.2 — LLM Models for Parsing, Matching & Monitoring

Model	Provider	Context	Best Use in Pipeline
Claude Haiku 4.5	Anthropic	1M tokens	High-volume real-time parsing; budget pipelines
Claude Sonnet 4.6	Anthropic	1M tokens	Best general-purpose agent (recommended for Agents 1, 2, 8)
Claude Opus 4.7	Anthropic	1M tokens	Complex agentic tasks; nuanced evaluation; pipeline monitoring
Gemini 3.5 Flash	Google	1M tokens	Fast real-time processing; multilingual workloads
GPT-5.5	OpenAI	1.05M tokens	Top reasoning; pipeline anomaly detection (Agent 8)
DeepSeek V4 Pro	DeepSeek	128K tokens	Budget flagship — near-frontier quality at 90% lower output cost
DeepSeek V4 Flash	DeepSeek	128K tokens	Ultra-budget high-volume tasks
Qwen 3.7 Max	Alibaba	1M tokens	Multilingual workloads; agentic tasks
Nemotron 30B	NVIDIA / OpenRouter	1M tokens	Current implementation — very low cost; large context

5.3 — Embedding Models for Semantic Search

Model	Dimensions	Context	Multilingual	Hosting
Voyage AI voyage-3-large	1024–2048	32K	Yes	API
Cohere embed-v4	1536	128K	Strong (100+)	API
OpenAI text-embedding-3-large	3072	8K	Moderate	API

Model	Dimensions	Context	Multilingual	Hosting
OpenAI text-embedding-3-small	1536	8K	Moderate	API (budget)
BGE-M3	1024	8K	100+ languages	Self-hosted
NV-Embed-v2	4096	32K	Moderate	Self-hosted
Nomic Embed v2	768	8K	Strong	Self-hosted
EmbeddingGemma-300M	768	8K	Yes	Self-hosted
Qwen3-Embedding	Variable	32K	100+ languages	Self-hosted

5.4 — Reranking Models

Model	Provider	Pricing	Best For
Cohere Rerank-3.5	Cohere	\$1.00 per 1,000 searches	End-to-end Cohere RAG pipelines; production
BGE-reranker-v2	BAAI	Free (self-hosted)	Open-source cross-encoder; lifts recall 10–25%
Voyage AI rerankers	Voyage AI	Free (first 200M tokens)	Pairs natively with Voyage embeddings

6. AI Model Price Comparison

All prices in USD per 1 million tokens as of June 2026. Prices verified from official provider pages and independent trackers. Open-source / self-hosted models have no API cost but require GPU infrastructure (estimated \$0.50–\$5.00/hour cloud GPU depending on model size).

6.1 — LLM Pricing Comparison

Model	Input \$/1M	Output \$/1M	Batch Discount	Notes
Claude Haiku 4.5	\$1.00	\$5.00	50% off	Cheapest Claude tier; high-volume parsing
Claude Sonnet 4.6	\$3.00	\$15.00	50% off	Best price/quality balance for production agents
Claude Opus 4.7	\$5.00	\$25.00	50% off	Premium agentic; complex evaluation tasks
Gemini 3.5 Flash	\$1.50	\$9.00	Available	Fast; good for real-time multilingual work

Model	Input \$/1M	Output \$/1M	Batch Discount	Notes
GPT-5.5	\$5.00	\$30.00	50% off	Strongest reasoning; 20% more expensive output than Opus
DeepSeek V4 Pro	\$0.44	\$0.87	Available	~90% cheaper output vs GPT-5.5; near-frontier quality
DeepSeek V4 Flash	\$0.14	\$0.28	Available	Ultra-budget; high-volume simple tasks
Qwen 3.7 Max	\$1.25	\$3.75	Available	Competitive multilingual; good agentic performance
Nemotron 30B (current)	\$0.05	\$0.20	—	Current implementation; cheapest API option

Key Pricing Takeaways

- DeepSeek V4 Pro at \$0.44/\$0.87 delivers near-frontier quality at roughly 90% less output cost than GPT-5.5 — strongest cost-performance case for parsing and matching agents
- Claude Sonnet 4.6 at \$3.00/\$15.00 is the recommended balance point for production agentic use
- Nemotron 30B at \$0.05/\$0.20 is the current implementation and remains one of the cheapest options for parsing at scale
- Batch API (50% off) is available for all Anthropic and OpenAI models — highly recommended for bulk upload processing (600 CVs) where latency is not critical

6.2 — Embedding Model Pricing

Model	Price \$/1M tokens	Dimensions	Context	Notes
Voyage AI voyage-3-large	\$0.18	1024–2048	32K	Highest retrieval quality; top MTEB scores
Cohere embed-v4	\$0.12	1536	128K	Best multilingual + hybrid search; longest context
OpenAI text-embedding-3-large	\$0.13	3072	8K	Industry standard; 3072-dim for high precision
OpenAI text-embedding-3-small	\$0.02	1536	8K	Budget option; sufficient for most RAG tasks
BGE-M3	Free (self-hosted)	1024	8K	Best open-source; 100+ languages; zero API cost
NV-Embed-v2	Free (self-hosted)	4096	32K	Topped MTEB benchmarks; NVIDIA Research
Nomic Embed v2	Free (self-hosted)	768	8K	Lightweight; strong multilingual; low RAM

Model	Price \$/1M tokens	Dimensions	Context	Notes
EmbeddingGemma-300M	Free (self-hosted)	768	8K	Runs in <200MB RAM; edge deployments
Qwen3-Embedding	Free (self-hosted)	Variable	32K	Instruction-aware; 100+ languages

Note: Embedding costs are negligible at Career141's scale. Indexing the full 115,000-candidate database with Voyage AI voyage-3-large (the most expensive option at \$0.18/M) would cost approximately \$4–\$10 for a one-time index build, assuming ~500 tokens per candidate profile.

6.3 — Monthly Cost Estimate for Career141

Based on a steady-state workload of 500 new CVs per week (2,000/month) and 50,000 candidate searches per month across the 115k+ database.

Scenario	CV Parsing (Agent 1)	Semantic Search (Agent 2)	LLM Matching	Est. Monthly Total
Budget — Nemotron 30B + BGE-M3 (self-hosted) + DeepSeek V4 Flash	~\$0.10	~\$0 (self-hosted)	~\$0.10	~\$0.20–\$2
Balanced — Sonnet 4.6 + OpenAI embed-3-small + Sonnet 4.6	~\$6	~\$1	~\$9	~\$16–\$20
Premium — Opus 4.7 + Voyage-3-large + Opus 4.7	~\$10	~\$9	~\$50	~\$70–\$80

Batch API (50% discount) is available for all Anthropic and OpenAI models and is strongly recommended for bulk upload processing, where real-time latency is not required. Applying batch processing to CV parsing alone can halve the parsing cost in the Balanced scenario.

7. Recommended Model Stacks by Deployment Type

Deployment Type	CV Parsing (Agent 1)	Embeddings (Agent 2)	LLM Agent	Reranking
High-Accuracy Enterprise	Google Document AI or LlamaParse	Voyage AI voyage-3-large	Claude Opus 4.7	BGE-reranker-v2
Cost-Optimised Production	Nemotron 30B or Claude Sonnet 4.6	BGE-M3 (self-hosted) or Cohere embed-v4	Claude Sonnet 4.6	Cohere Rerank-3.5

Deployment Type	CV Parsing (Agent 1)	Embeddings (Agent 2)	LLM Agent	Reranking
High-Volume / Real-Time	Gemini 3.5 Flash or Claude Haiku 4.5	EmbeddingGemma-300M or Nomic Embed v2	Gemini 3.5 Flash or DeepSeek V4 Pro	BGE-reranker-v2
Multilingual Deployment	Nemotron 30B or Gemini 3.5 Flash	BGE-M3 or Qwen3-Embedding (100+ langs)	Claude Opus 4.7 or Gemini 3.5 Flash	Cohere Rerank-3.5

8. AI Model Recommendations by Agent

The table below maps each of the 8 AI agents to their recommended model options, from current implementation to enterprise upgrades.

Agent 1 — CV Parsing & AI Extraction

Vision Language Models (Document Understanding)	Language Models (Field Extraction)
Google Document AI — enterprise-grade, Gemini-powered	Claude Opus 4.7 — optimised for extended agentic tasks
LlamaIndex LlamaParse — tables, charts, multi-column layouts	GPT-5.5 — superior reasoning and structured output
Doxis AI.dp — >99% accuracy, GDPR compliant	Gemini 3.5 Flash — fast real-time agentic processing
ABBYY FlexiCapture — large enterprise capture	DeepSeek V4 Pro — 75% cheaper than frontier models
Amazon Textract — AWS-native, multi-language	Qwen 3.7 Max — strong multilingual support

Current Implementation: OpenRouter Nemotron 30B (lightweight, cost-effective, temperature=0)

Agent 2 — Candidate Matching & Semantic Search

Embedding Models	Reranking & Matching LLMs
Voyage AI voyage-3-large — highest retrieval quality	BGE-reranker-v2 — cross-encoder; lifts recall 10–25%
OpenAI text-embedding-3-large — industry standard	Cohere Rerank-3.5 — integrated with Cohere pipelines
Cohere embed-v4 — hybrid search support	

BGE-M3 — 100+ languages, self-hosted	Claude Opus 4.7 — nuanced candidate evaluation
Qwen3-Embedding — instruction-aware, 100+ langs	Claude Sonnet 4.6 — cost-efficient high-volume matching
EmbeddingGemma-300M — <200MB RAM, edge use	Gemini 3.5 Flash — real-time scoring
NV-Embed-v2 — topped MTEB benchmarks	
Nomic Embed v2 — strong multilingual, low cost	

Agent 6 — Deduplication & Profile Creation

Recommended models for intelligent deduplication and profile merging:

- OpenAI text-embedding-3-large
- BGE-M3
- Claude Opus 4.7

Stack: Vector embeddings (semantic similarity) + SHA-256 (exact match) + LLM validation (edge cases)

Agent 8 — Pipeline Health Agent

Recommended models for daily pipeline monitoring and anomaly detection:

- GPT-5.5 Pro
- Claude Opus 4.7
- Gemini 3.5 Flash

9. Summary

The Career141 CV Processing & Recruitment Pipeline is a production-grade, AI-first system that automates the most time-consuming parts of the recruitment process while preserving human judgment at the decision points that matter most.

What This System Delivers

- 7 ingestion channels feeding one unified pipeline — no candidate slips through the cracks
- 8 AI agents running autonomously 24/7 — monitoring inboxes, parsing CVs, scoring candidates, and running outreach
- 115,000+ candidate profiles indexed for semantic search — every historical CV is a potential future placement
- Three-level approval workflow (TA → Director → Client) — quality control at every stage with full audit trail
- Monthly AI infrastructure cost as low as \$2 (budget stack) or ~\$20 (balanced production stack)
- Proactive AI outreach covers both new applicants and database candidates — email follow-up sequences (Agent 3) and AI phone calls (Agent 5)
- All candidate history is permanent and searchable — rejected candidates today become placed candidates tomorrow

This proposal represents the complete technical and operational foundation for the Career141 platform, documented from the live Miro system diagrams and the development team's implementation notes.

Version 1.0 | June 2026 | Career141 Internal